

Finite-Source Cosmology from Observer-Patch Holography

Source, Transfer, and Likelihood Boundaries

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Abstract

We give a conditional source-screen construction for Observer-Patch Holography (OPH). A supplied primitive collar law fixes the angular amplitude, and an edge-center generator with its orientation-half identity fixes the tilt once its reserve-generator receipt is supplied. Physical dilation and radial cross-covariance tomography provide two sufficient uniqueness routes beyond one-shell data. The construction states the source, covariant stress, physical scale, Boltzmann transfer and likelihood conditions needed to convert these screen quantities into cosmological observables. On the sourced causal order the expanding background is a record density and the redshift is a fourth root of a count ratio. The receipt-conditional primordial values, the tilt $1 - P_\star/48 = 0.96602$, zero running, zero source-side isocurvature and zero primordial tensor amplitude, are compared with Planck 2018, ACT DR6, SPT-3G and BICEP/Keck data in a deterministic ledger with machine-checked brackets: the tilt lies within one standard deviation of Planck 2018 and of both SPT-3G combinations and 2.4 standard deviations below the ACT DR6 combination with lensing and baryon acoustic oscillations.

1 Claim Boundary

A physical cosmological prediction requires one construction that maps finite OPH records to a covariant source, propagates that source through a declared background, and evaluates the resulting observables with a complete likelihood. No such construction is supplied. Simulator outputs may test numerical interfaces and expose incompatible assumptions. They carry no cosmological verdict.

The same source premises govern the dark-sector, cosmological-vacuum, and compact OPH constructions.

2 Finite-Source Interface

Let X_r denote a finite observer-screen state at refinement level r . A cosmological source map must emit

$$S_r(X_r) = (g_{\mu\nu}, T_{\mu\nu}, \mathcal{I}, \mathcal{B})_r, \quad (1)$$

where $T_{\mu\nu}$ is conserved, $\mathcal{I}$ contains gauge-invariant initial data, and $\mathcal{B}$ states the background and boundary conditions. The map must be defined without using the observed sky quantities that its outputs are compared with.

The complete one-shell covariance determines the three-dimensional correlation only for chord lengths $0 \leq s \leq 2R_\star$. The resulting radial kernel is infinite-dimensional and persists under positivity. Two sufficient routes for a source-derived lift are a scale-natural embedding that transports refinement to physical log-wavenumber dilation and complete radial cross-covariances supporting spherical-Hankel tomography. Each route has its own hypotheses. The physical source map must also specify the radial measure, null-space treatment, clock, normalization, and non-fitting forward residual. No finite source construction satisfying these conditions is supplied here. Its source-dependency graph must be finite and acyclic. This graph is a scientific prerequisite graph among stress, clock, scale, radial, transfer, and likelihood objects. It is distinct from the semantic read-from order of the consensus layer. If its nodes are instantiated by semantic commits, a declared mapping and complete certified read supports can make the generated informational order verify their execution dependencies. The provenance theorem alone derives neither these scientific nodes nor their prerequisite edges. The source Gram construction supplies a three-dimensional spatial carrier and, after adjoining a real axis, an algebraic Lorentzian carrier. A nonzero closed convex pointed displacement cone is forced to be future or past Lorentzian if the proper icosahedral rotations and all rapidities of one boost axis preserve it. This is an operational condition on histories, interventions and clock readouts; an algebraic frame action alone does not satisfy it. A time orientation fixes the future branch [1].

A conservative Fibonacci-record family realizes a controlled flat $1 + 3$ limit through a specified population, complete shrinking-radius neighbour reads and layer duration. Its normalized event counts converge to spacetime volume, and interior timelike intervals have ordering fraction tending to $1/10$. Given complete authenticated ancestry and interval anchors, an observer can read $(|I|/|J|)^{1/4}$ relative to a nondegenerate interval J in the same history. It converges to the proper-time ratio without using coordinates, timestamps or density in the readout. The population and local read law remain hypotheses of the limit, and the reference interval fixes only a relative time unit [1].

Cosmology requires a common physical realization of those events and clocks with expanding geometry, conserved stress, a primordial source, a radial lift and photon transfer. The finite tensor theorem uses a separately supplied $3 + 1$ tensor interface, nine algebraic null balances, symmetry, discrete conservation and connectedness to obtain an Einstein-shaped identity with one constant metric term. It does not identify its tensor-coordinate differences with source-event displacements. Smooth Einstein promotion requires same-family tensor-curvature convergence or the continuum small-ball/null-balance identification; scalar-curvature convergence alone is diagnostic. The flat count-clock construction does not supply these cosmological attachments or a likelihood.

2.1 Expansion on the sourced order

The transfer calculation of Section 6 runs on a spatially flat Friedmann background. On the source family that background has an exact finite form: the layered read law in comoving coordinates is conformally invariant, so the flat causal order of the spacetime construction is the causal order of every spatially flat expanding history, and the scale factor enters only through the count measure, as the weight a^4 per comoving cell and conformal tick that a constant physical event density assigns. The count clock then reads proper time, and the redshift between two epochs is the fourth root

of the ratio of record counts of congruent comoving diamonds. The statements are proved in the fragment below and machine-checked in the named Lean module. The record-production profile $a(\eta)$ is a supplied datum on this family; deriving it from the source dynamics is the sourced-density problem of the spacetime paper, and nothing here supplies the Friedmann equation, a physical clock, or the physical scale map.

2.2 Expansion as record density on the source order

The spatially flat Friedmann–Lemaître–Robertson–Walker metric in conformal time η is

$$ds^2 = \sigma(\eta)^2 (c^2 d\eta^2 - |dx|^2), \quad (2)$$

with scale factor $\sigma(\eta) > 0$; the letter a is the edge radius of the read law. This metric is a positive conformal factor times the flat metric in the coordinates (η, x) . A positive conformal factor leaves the causal order of a Lorentzian metric unchanged, and the causal order fixes the metric up to such a factor (Hawking, King and McCarthy 1976; Malament 1977). On the spatially flat family the conformal factor is the entire background, so every profile σ has the causal order of flat space in comoving coordinates. The layered read law of the source-net construction places an event at every (j, s) for a site s in a population S , lets $(j + 1, t)$ read (j, s) whenever $\|t - s\| \leq a$, and takes the reflexive transitive closure of the reads as the order. Written in comoving coordinates with the tick $\Delta = a/c$ in conformal time, this law generates one finite order for every profile. The profile enters only through the count measure.

Proposition 2.1 (Conformal invariance of the layered order). *Let S be a subset of a real normed space, $a > 0$ an edge radius and $\sigma_j > 0$ the scale at layer j . The physical position of the site s at layer j is $\sigma_j s$ and the physical read radius at that layer is $\sigma_j a$. The physical read from (j, s) to $(j + 1, t)$,*

$$\|\sigma_{j+1}t - \sigma_{j+1}s\| \leq \sigma_{j+1}a,$$

holds exactly when $\|t - s\| \leq a$. The layered orders generated by the physical reads and by the comoving reads coincide, for every profile $(\sigma_j)_{j \geq 0}$.

Proof. For $\sigma > 0$, $\|\sigma t - \sigma s\| = \sigma \|t - s\|$, and multiplication by σ preserves and reflects the comparison with σa . The two read relations are equal, hence so are their reflexive transitive closures. $\square$

Proposition 2.2 (FLRW count measure). *Assign a comoving cell of volume v_s to every site s and the conformal tick Δ to every layer. Under a constant physical event density ρ , the physical tick at layer j is $\sigma_j \Delta$ and the physical cell volume is $\sigma_j^3 v_s$, so a finite set I of events carries the mass*

$$M_\sigma(I) = \rho \sum_{(j,s) \in I} (\sigma_j \Delta) (\sigma_j^3 v_s) = \rho \sum_{(j,s) \in I} \sigma_j^4 \Delta v_s, \quad (3)$$

the discrete form of the four-volume weight $\sigma^4 d\eta d^3x$. Write $M_1(I) = \rho \Delta \sum_{(j,s) \in I} v_s$ for the flat mass. If $0 \leq \sigma_{\min} \leq \sigma_j \leq \sigma_{\max}$ on the layers of I , then

$$\sigma_{\min}^4 M_1(I) \leq M_\sigma(I) \leq \sigma_{\max}^4 M_1(I). \quad (4)$$

A constant profile $\sigma_j = \sigma$ gives $M_\sigma(I) = \sigma^4 M_1(I)$.

Proof. The identity in (3) is the product of the two scaled factors. Fourth powers are monotone on nonnegative reals and $\Delta v_s \geq 0$, so each term of the sum lies between the same term with σ_j replaced by $\sigma_{\min}$ and by $\sigma_{\max}$. Summing proves (4); the constant case is the common factor σ^4 . $\square$

Corollary 2.3 (Count clock in an expanding family). *Let I and J be finite event sets with uniform cells, J nonempty, with $\sigma_{\min}^I \leq \sigma_j \leq \sigma_{\max}^I$ on the layers of I , $\sigma_{\min}^J \leq \sigma_j \leq \sigma_{\max}^J$ on the layers of J , $\sigma_{\min}^I \geq 0$ and $\sigma_{\min}^J > 0$. Then*

$$\frac{\sigma_{\min}^I}{\sigma_{\max}^J} \left(\frac{|I|}{|J|} \right)^{1/4} \leq \left(\frac{M_\sigma(I)}{M_\sigma(J)} \right)^{1/4} \leq \frac{\sigma_{\max}^I}{\sigma_{\min}^J} \left(\frac{|I|}{|J|} \right)^{1/4}. \quad (5)$$

Let I and J approximate comoving diamonds of conformal durations T_I and T_J in the sense of the count-clock theorem for the flat order, so that $(|I|/|J|)^{1/4} \rightarrow T_I/T_J$ under refinement. A comoving worldline through a diamond of conformal duration T accrues proper time between $\sigma_{\min}T$ and $\sigma_{\max}T$. In the refinement limit the clock read from the physical masses, $(M_\sigma(I)/M_\sigma(J))^{1/4}$, and the proper-time ratio τ_I/τ_J lie in one interval whose endpoints have the quotient $(\sigma_{\max}^I/\sigma_{\min}^I)(\sigma_{\max}^J/\sigma_{\min}^J)$; the two readings agree up to this factor in either direction. For a profile continuous in η and diamonds shrinking to points, the factor tends to one and the clock reads the proper-time ratio exactly in the limit.

Proof. Divide the two sandwich inequalities (4) for I and J , with $M_1(I)/M_1(J) = |I|/|J|$ for uniform cells, and take the increasing fourth root; the fourth root of σ^4 is σ . The proper time along the comoving worldline is $\int \sigma d\eta$ over the conformal duration, bounded by the extreme scales times T ; the ratio of two such enclosures has the stated endpoint quotient. Continuity of σ forces $\sigma_{\max}/\sigma_{\min} \rightarrow 1$ on shrinking diamonds. $\square$

Corollary 2.4 (Redshift as a count ratio). *Let two congruent comoving diamonds, with the same comoving cells and the same conformal ticks, lie at the emission epoch, of scale σ_e , and at the reception epoch, of scale σ_0 , the profile being constant across each. Their masses are $n_e = \kappa \sigma_e^4$ and $n_0 = \kappa \sigma_0^4$ with the common factor $\kappa = \rho \Delta \sum_{(j,s) \in I} v_s$, and*

$$1 + z = \frac{\sigma_0}{\sigma_e} = \left(\frac{n_0}{n_e} \right)^{1/4}. \quad (6)$$

Proof. The constant case of Proposition 2.2 gives both masses; κ cancels in the ratio, and the positive fourth root of $(\sigma_0/\sigma_e)^4$ is σ_0/σ_e . $\square$

Remark 2.5 (Expansion history as a record-production law). On this family the Friedmann equation is a statement about record production: a comoving cell of volume v produces $\rho \sigma(\eta)^4 \Delta v$ records per conformal tick, and a profile $\sigma(\eta)$ is the same datum as a record-production rate per comoving cell. The de Sitter profile $\sigma(\eta) = -1/(H\eta)$ on $\eta < 0$ solves $d\sigma/d\eta = H\sigma^2$, and its record density per comoving cell is $n(\eta) \propto (H\eta)^{-4}$; the fixed-capacity branch of the corpus, $w = -1$, corresponds to this profile. The Lean module `Lean/Geometry/SourceNetConformalRecordDensity.lean` checks the order invariance, the mass identity and sandwich, the clock enclosure (5), the redshift identity and the de Sitter derivative and density identities over the reals.

Three items lie outside this construction. The source law selecting the profile, the record-production rate per comoving cell, is not derived; the profile is a supplied datum. A reweighted flat order determines at most the conformal factor; on the spatially flat family that factor is the entire background, and spatial curvature $k \neq 0$ changes the comoving spatial metric, which this construction does not treat. No physical clock and no population selection by native repair follow from these identities.

2.3 Conditional source spectrum

On a source-stress-complete uniform-density release cut, the screen field is

$$q_r = \Pi_{\geq 2,r} \left[\frac{1}{3} \log \frac{J_{X,r}}{J_{X,r}} \right]. \quad (7)$$

The primitive collar ledger selects a relative-MaxEnt release law before the screen energy is evaluated. It excludes spectra, residuals, likelihoods, fitted amplitudes, and measurement-calibrated proxies. The release energy then fixes

$$A_{q,r} = \frac{1}{d_r} \mathbb{E}_{\nu_r^{\text{rel}}} [q_r^\top M_r K_{\theta,r} q_r] = \frac{2E_{q,r}^{\text{src}}}{d_r}. \quad (8)$$

The conformal precision and angular spectrum are

$$K_\theta = \frac{\Gamma(\sqrt{-\Delta_{S^2} + 1/4} + 3/2 + \theta/2)}{\Gamma(\sqrt{-\Delta_{S^2} + 1/4} - 1/2 - \theta/2)}, \quad (9)$$

$$C_\ell^q = A_q \frac{\Gamma(\ell - \theta/2)}{\Gamma(\ell + 2 + \theta/2)}, \quad \ell \geq 2. \quad (10)$$

A required infinitesimal reserve-generator receipt must supply a full-collar density $P_\star/24$. A separately proved weighted orientation-reversal coarea identity then supplies its source-facing half, so

$$\theta = \frac{P_\star}{48}, \quad n_s = 1 - \frac{P_\star}{48}, \quad \kappa_{\text{rep}}^{\text{edge}} = \frac{P_\star}{48(P_\star - \varphi)}. \quad (11)$$

A finite one-step survival probability has exponent $-\log u_q(\log b)/\log b$ and does not supply this infinitesimal density.

On the source-dilation branch, the finite source embedding must satisfy

$$D_{s,r} U_r = U_r R_{s,r}, \quad C_{\zeta,r} = U_r C_{\text{src},r} U_r^*. \quad (12)$$

Strong finite-to-continuum convergence, a uniform covariance norm bound, and a vanishing operator residual then give

$$D_s^{-1} C_\zeta D_s = e^{-\theta s} C_\zeta. \quad (13)$$

This identity forces

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}, \quad A_\zeta = \frac{A_q}{\pi^{3/2} Z_q^2 (k_\star R_\star)^\theta} \frac{\Gamma(3/2 + \theta/2)}{\Gamma(1 + \theta/2)}. \quad (14)$$

The tomography branch obtains uniqueness from complete radial cross-covariances. A finite prior selects a conditional radial continuation. Every finite branch publishes its singular spectrum, null basis, finite-window error, positivity check, and forward residual. Temperature and polarization spectra belong to the transfer interface.

3 Dark-Sector Interface

The dark-matter paper proposes anomalous modular charge on overlap collars. For a conserved total charge Q_A in a comoving region of physical volume $V_0 a^3$, its homogeneous density satisfies

$$\rho_A = \frac{Q_A}{V_0 a^3} \propto a^{-3}. \quad (15)$$

Under the separate homogeneous continuity equation with no exchange, this dilution is the pressureless background law. Deep-regime scale covariance and quadrature composition characterize the static spherical acceleration law and baryonic Tully–Fisher scaling.

Equation (15) does not determine the abundance, initial perturbations, relativistic stress, gravitational slip, sound speed, nonlinear evolution, or coupling to the visible sector. A cosmological use requires a covariant completion that supplies these quantities from one common source and obeys the full energy-momentum ledger. That completion is not supplied here.

4 Primordial Interface

A physical primordial branch must supply all of the following:

1. a finite source ensemble and a clock;
2. a conserved covariant stress tensor;
3. gauge-invariant scalar, vector, and tensor initial data;
4. a screen-to-volume lift with a controlled null space;
5. phase, amplitude, and isocurvature conditions;
6. refinement stability and an error model.

The conditional source theorem selects $n_s = 1 - P_\star/48$ from the infinitesimal edge-center generator receipt. Physical primordial use requires one finite source construction that supplies this receipt together with the release law, stress, clock, freezeout, mode, scale, and radial receipts. Numerical agreement with a measured summary does not supply the missing source map.

5 Comparison with Public Data

The values that the conditional theorems of this paper and its companions fix without a dial are compared with public measurements in a deterministic ledger released with these papers, with an independent verifier and exact rational brackets machine-checked in the Lean module `CosmologyLedgerBrackets`. Every comparison is on seen data and carries a class label; the rule, theory minus measurement in units of the quoted standard deviation, is fixed before any row is read. The primordial rows are the following.

- The tilt $n_s = 1 - P_\star/48 = 0.96602$ at the comparison pixel, with a branch spread below 10^{-7} , lies 0.27σ above Planck 2018 (0.9649 ± 0.0042), 0.69σ above SPT-3G with Planck (0.9636 ± 0.0035), 0.79σ below the SPT, Planck and ACT combination (0.9684 ± 0.0030), 1.28σ below ACT DR6 with Planck (0.9709 ± 0.0038) and 2.43σ below ACT DR6 with Planck, lensing and baryon acoustic oscillations (0.9743 ± 0.0034). The value is a consequence of the reserve-generator receipt, which is not supplied; the comparison records where a receipt-conditional number lands and selects no branch.

- The running vanishes exactly on the dilation-cocycle branch; Planck 2018 gives -0.0045 ± 0.0067 (0.67σ) and ACT DR6 with Planck, lensing and baryon acoustic oscillations gives 0.0062 ± 0.0052 (1.19σ).
- The primordial tensor amplitude vanishes exactly, since the source is the scalar collar-volume field and no transverse-traceless source is emitted; BICEP/Keck 2018 gives $r_{0.05} < 0.036$ and the fourth Planck release with BK18 and BAO gives $r < 0.032$. Generic slow-roll inflation gives $r > 0$, so this row discriminates, and a null at the LiteBIRD design sensitivity below 0.001 would discriminate against the models it keeps alive.
- The source-side isocurvature fraction vanishes on the rank-one single-clock branch; Planck 2018 bounds the uncorrelated cold-dark-matter fraction at 3.5, 3.8 and 3.9 percent at the three reference scales. Zero non-Gaussianity at the source (Planck local -0.9 ± 5.1) and zero curvature (Planck with BAO 0.0007 ± 0.0019 ; DESI DR2 with CMB 0.0023 ± 0.0011 , 2.1σ) are shared with the adiabatic single-field baseline.

The background rows are the fixed-capacity equation of state $(w_0, w_a) = (-1, 0)$, which the Gaussian moment diagnostic of the official DESI DR2 chains places between 2.6σ and 3.9σ from the posterior means of the four default combinations, a tension recorded with the same prominence as the agreements, and the capacity candidates 3.29×10^{122} and 3.30×10^{122} , whose dictionary $\Lambda \ell_P^2 = 3\pi/N$ returns the cosmological constant within 0.6 percent of the Planck-central value and 2.7σ below the DESI DR2 with CMB display. The primordial registration proposal released with the ledger states verbatim-ready freeze rows for the tilt, the tensor zero, and the running and isocurvature zeros, with every numeric kill cell an owner slot, so that the next primary-CMB and B-mode releases can score them.

6 Transfer Interface

Given a completed source map, the transfer calculation must declare the background species, equations of state, collision terms, recombination model, initial modes, gauge convention, numerical tolerances, and observable projection. Standard Einstein–Boltzmann software can test this interface with imported inputs. Such a run tests the solver path rather than an OPH source theorem.

The minimum transfer outputs are

$$(C_\ell^{TT}, C_\ell^{TE}, C_\ell^{EE}, C_L^{\phi\phi}, P_m(k, z), H(z), D_A(z), f\sigma_8(z)). \quad (16)$$

All outputs must arise from one compatible background and perturbation system.

7 Likelihood Interface

A data comparison must state the datasets, masks, covariances, nuisance parameters, priors, and combination rule. Overlapping samples and shared calibrations must be represented in the covariance model. Visual overlays, diagonal residual sums, and comparisons made after selecting a formula are diagnostics.

The cosmic microwave background (CMB), galaxy, and background comparisons are diagnostic. These computations supply neither a prospectively fixed cosmological comparison nor a falsification verdict.

8 Theorem scope

Object	Classification
Finite observer-screen carriers	Conditional finite construction
Screen scalar covariance	Conditional theorem
Source-functional screen amplitude and edge-center tilt	Conditional theorem; finite receipt not supplied
Radial-lift mathematics	Conditional theorem packet; one-shell unrestricted inversion is impossible
Physical source-dilation or radial-tomography receipt	Not supplied
Primordial covariant source	Not supplied
Repair-charge cosmological abundance	Not supplied
Relativistic repair stress and slip	Not supplied
Boltzmann initial-state bridge	Not supplied
Joint CMB, baryon acoustic oscillation, lensing, growth, and cluster likelihood	Not supplied
Prospectively fixed cosmological comparison	Not supplied by these diagnostics

9 Conclusion

Finite OPH screen data become cosmology only through a covariant source map, a three-dimensional lift, a conserved stress tensor, compatible transfer equations, and a complete likelihood. These constructions are not supplied here. Existing cosmological calculations are diagnostics and carry no falsification verdict.

References

- [1] B. Müller et al., *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.tex.

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